

1. Dividing a program into functions

- a. is the key to object-oriented programming.
- b. makes the program easier to conceptualize.
- c. may reduce the size of the program.
- d. makes the program run faster.

Ans: b. makes the program easier to conceptualize.

2. A function name must be followed by parentheses ().

3. A function body is delimited by Curly braces { }.

4. Why is the main() function special?

Ans: The main() function is special function because it is the first executed when the program starts

5. A C++ instruction that tells the computer to do something is called a Statement.

6. Write an example of a normal C++ comment and an example of an old-fashioned /* comment.

Ans: // This is a Normal C++ comment

/* This is a
Multiline (old fashioned) comment */

7. An expression

- a. usually evaluates to a numerical value.
- b. indicates the emotional state of the program.
- c. always occurs outside a function.
- d. may be part of a statement.

Ans: a. usually evaluates to a numerical value.

8. Specify how many bytes are occupied by the following data types in a 32-bit system:

- a. Type int
- b. Type long double
- c. Type float
- d. Type long

Ans: a. 4 bytes b.10 bytes c. 4 bytes d. 4 bytes

9. True or false: A variable of type char can hold the value 301.

Ans: False

10. What kind of program elements are the following?

- a. 12
- b. 'a'
- c. 4.28915
- d. JungleJim
- e. JungleJim()

**Ans: a. integer constant b. Character constant c. Floating constant
d. Variable/Function name e. Function call**

11. Write statements that display on the screen

- a. the character 'x'
- b. the name Jim
- c. the number 509

Ans: a. cout<<'x'; b. cout<<"jim"; c. cout<<509;

12. True or false: In an assignment statement, the value on the left of the equal sign is always equal to the value on the right.

Ans: False

13. Write a statement that displays the variable george in a field 10 characters wide.

Ans: cout << setw(10) << george;

14. What header file must you #include with your source file to use cout and cin?

Ans: #include<iostream>

15. Write a statement that gets a numerical value from the keyboard and places it in the variable temp.

Ans: cin >> temp;

16. What header file must you #include with your program to use setw?

Ans: #include<iomanip>

17. Two exceptions to the rule that the compiler ignores whitespace are String constant and preprocessor directives.

18. True or false: It's perfectly all right to use variables of different data types in the same arithmetic expression.

Ans: True

19. The expression 11%3 evaluates to 2.

20. An arithmetic assignment operator combines the effect of what two operators?

Ans: Arithmetic operator and Assignment operator

21. Write a statement that uses an arithmetic assignment operator to increase the value of the variable temp by 23. Write the same statement without the arithmetic assignment operator.

Ans: `int temp;`
`temp += 23; // using arithmetic assignment operator`
`temp = temp + 23; // without arithmetic assignment operator`

22. The increment operator increases the value of a variable by how much?

Ans: The increment operator increase the value of a variable by 1

23. Assuming var1 starts with the value 20, what will the following code fragment print out? `cout << var1--;` `cout << ++var1;`

Ans: `int var1 = 20;`
`cout << var1--;`
`cout << ++var1;`
Output is: 2020

24. In the examples we've seen so far, header files have been used for what purpose?

Ans: To provide declarations (prototypes) for library functions and objects, such as `cout`, `cin`, and `setw`

25. The actual code for library functions is contained in a Library file.